

InsightGPT Lite

Analytics Report

AI-Powered Data Analytics and Retrieval-Augmented Query Platform

Internship Project - BCA Data Science

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1. Project Information

InsightGPT Lite is an AI-powered data analytics and retrieval-augmented query platform designed to analyze uploaded datasets and generate actionable insights.

2. Dataset Overview

Total Rows: 649

Total Columns: 33

Numerical Columns: 16

Categorical Columns: 17

3. Data Quality Analysis

Total Missing Values: 0

Overall Missing Cell Percentage: 0.0%

Duplicate Records: 0

Missing Data Status: Low amount of missing data

Duplicate Status: No duplicate records detected

4. Column Information

Columns in Dataset

- school
- sex
- age
- address
- famsize
- Pstatus
- Medu
- Fedu
- Mjob
- Fjob
- reason
- guardian
- traveltime
- studytime
- failures
- schoolsup

- famsup
- paid
- activities
- nursery
- higher
- internet
- romantic
- famrel
- freetime
- goout
- Dalc
- Walc
- health
- absences
- G1
- G2
- G3

Column Data Types

school: str

sex: str

age: int64

address: str

famsize: str

Pstatus: str

Medu: int64

Fedu: int64

Mjob: str

Fjob: str

reason: str

guardian: str

traveltime: int64

studytime: int64

failures: int64

schoolsup: str

famsup: str

paid: str

activities: str
nursery: str
higher: str
internet: str
romantic: str
famrel: int64
freetime: int64
goout: int64
Dalc: int64
Walc: int64
health: int64
absences: int64
G1: int64
G2: int64
G3: int64

5. Statistical Summary

age: Mean=16.744, Median=17.000, Min=15.000, Max=22.000
Medu: Mean=2.515, Median=2.000, Min=0.000, Max=4.000
Fedu: Mean=2.307, Median=2.000, Min=0.000, Max=4.000
traveltime: Mean=1.569, Median=1.000, Min=1.000, Max=4.000
studytime: Mean=1.931, Median=2.000, Min=1.000, Max=4.000
failures: Mean=0.222, Median=0.000, Min=0.000, Max=3.000
famrel: Mean=3.931, Median=4.000, Min=1.000, Max=5.000
freetime: Mean=3.180, Median=3.000, Min=1.000, Max=5.000
goout: Mean=3.185, Median=3.000, Min=1.000, Max=5.000
Dalc: Mean=1.502, Median=1.000, Min=1.000, Max=5.000
Walc: Mean=2.280, Median=2.000, Min=1.000, Max=5.000
health: Mean=3.536, Median=4.000, Min=1.000, Max=5.000
absences: Mean=3.659, Median=2.000, Min=0.000, Max=32.000
G1: Mean=11.399, Median=11.000, Min=0.000, Max=19.000
G2: Mean=11.570, Median=11.000, Min=0.000, Max=19.000
G3: Mean=11.906, Median=12.000, Min=0.000, Max=19.000

6. Outlier Analysis

Column with Most Potential Outliers: failures

Potential Outliers: 100

The column with the highest number of potential outliers is 'failures', with 100 detected outliers.

7. Correlation Analysis

Strongest Correlation: G2 and G3

Correlation Coefficient: 0.919

The strongest numerical relationship is between 'G2' and 'G3', with a correlation coefficient of 0.919.

8. Automatic Dataset Insights

Dataset contains 649 rows and 33 columns.

Numerical columns: 16

Categorical columns: 17

The dataset contains no missing values.

No duplicate records were detected.

The column with the highest number of potential outliers is 'failures', with 100 detected outliers.

The strongest numerical relationship is between 'G2' and 'G3', with a correlation coefficient of 0.919.

9. Dataset Health

Health Score: 80/100

Overall Status: Good

Missing Data Impact: -5

Duplicate Impact: -0

Outlier Impact: -15

10. Recommended Actions

- Investigate potential outliers in 'failures' before performing advanced statistical or machine-learning analysis.
- Further investigate the relationship between 'G2' and 'G3'.

11. AI Executive Summary

Executive Summary

The dataset, comprising 649 rows and 33 columns (16 numerical, 17 categorical), demonstrates excellent data quality with no missing values or duplicate records identified. Initial analysis highlights a significant number of potential outliers within the 'failures' column, warranting further investigation. A very strong positive correlation of 0.919 was also observed between the 'G2' and 'G3' columns, indicating a close relationship between these variables.

Key Observations

- The dataset is comprehensive, containing 649 rows and 33 columns, with a balanced distribution of numerical and categorical data types.
- Data quality is exceptional, featuring zero missing values and no duplicate records across all columns.
- The 'failures' column contains 100 potential outliers, representing a notable subset of the data that requires closer examination.
- A strong positive correlation of 0.919 exists between the 'G2' and 'G3' columns.

Recommended Attention

- Investigate the 100 potential outliers identified in the 'failures' column to understand their nature and impact on overall analysis.
- Further explore the strong correlation between 'G2' and 'G3' to uncover the underlying relationship and its implications.
- Leverage the high data quality for robust and reliable insights in subsequent analytical efforts.

12. Report Details

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Platform: AI-Powered Data Analytics and RAG System